

Let's Practice Together: Exploring Collaborative Virtual Reality for Nursing Students' IVPB Medication Administration Training

¹Yupei Duan, ¹Xinhao Xu, ²Jhon Bueno-Vesga, ¹Shangman Li, ²Michelle Cox, ¹Yuanyuan Gu, ³Sue Yun Fowler, ³Hillary L. Claunch, and ⁴Jason Snyder

¹School of Information Science & Learning Technologies, University of Missouri, Columbia, MO, United States

²College of Information Sciences and Technology, The Pennsylvania State University, PA, United States

³Sinclair School of Nursing, University of Missouri, Columbia, MO, United States

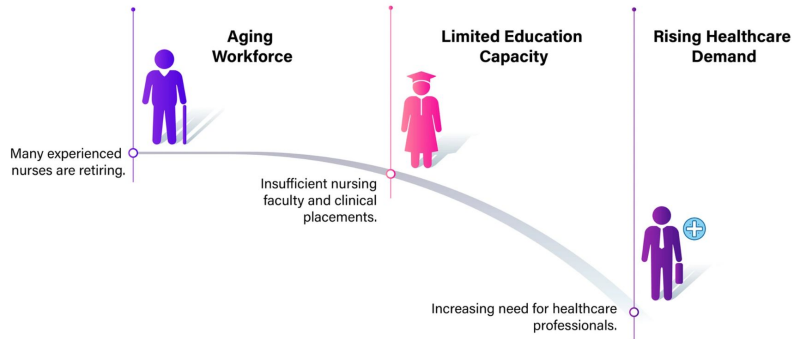
⁴Shelden Clinical Simulation Center, University of Missouri, Columbia, MO, United States



School of Information Science
& Learning Technologies
University of Missouri

Study Motivation

Factors Contributing to the Nursing Shortage



- WHO projects a global shortage of 4.5 million nurses by 2030.
- In the U.S., demand for nurses is rising due to an aging population, shortage of limited education capacity, and increasing healthcare demand.

Fig. 1 Cogent Marketing. (2025, September 16). *Nurse demographics and workforce statistics in the United States: Trends, diversity, and the growing shortage*. CWS Health.

<https://www.cwshealth.com/post/nurse-demographics-and-workforce-statistics-in-the-united-states-trends-diversity-and-the-growing>



Problem Context



Fig. 2 Real IV Pump Fig. 3 VR IV Pump

- IVPB medication administration is complex and safety-critical.
- Errors in this procedure can seriously affect patient safety.
- Traditional training may not provide enough safe and repeatable practice.
- VR offers immersive, risk-free practice opportunities.
- Collaborative VR for IVPB Medication Administration is still underexplored.



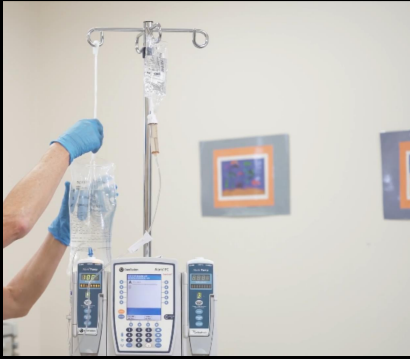


Fig. 4 Real IVPB manipulation



Fig. 5 VR IVPB manipulation

Purpose and Research Questions

- This pilot study explored how nursing students experienced collaborative practice in a VR-IVPB Medication Administration system.
- RQ1. How do nursing students rate the quality of their collaborative practice within the VR-IVPB Medication Administration system?
- RQ2. How do nursing students describe the impact of collaborative practice in the VR-IVPB Medication Administration system on their learning experience?



Theoretical Foundation

Social Interdependence Theory

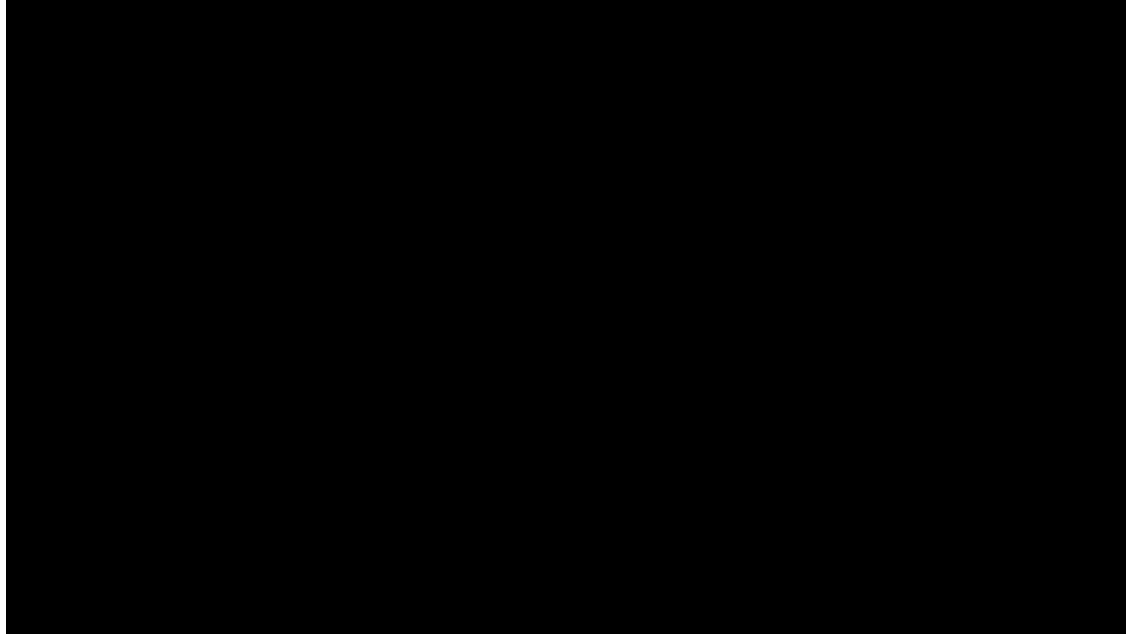
- Collaboration supports positive interdependence
- Peer interaction can promote support, accountability, and shared learning

Situated Learning Theory

- Learning is stronger in realistic, authentic practice contexts
- Scaffolding and immediate feedback support skill development



VR-IVPB Medication Administration System



School of Information Science
& Learning Technologies
University of Missouri

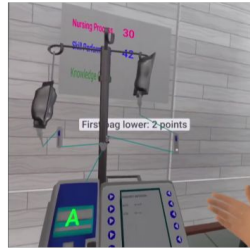
VR-IVPB Medication Administration System



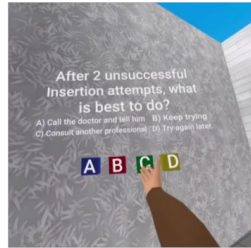
(a)



(b)



(c)



(d)

Fig. 6 Features of VR-IVPB Medication Administration System: (a) Overview of the training scenario; (b) Interactive IV Pump Operation; (c) Real-time Performance Scoring; (d) Integrated Knowledge Assessment

- Built in Meta Horizon Worlds
- Students practiced as 3D avatars
- Supported both individual and collaborative practice
- Simulated key IVPB medication administration equipment and procedures, including *PPE*, *roller clamps*, *drip chambers*, *IV pumps*, *tubing connectors*, *IV bag hangers*, *piggybacks*, *patient insert spikes*, etc.
- Included embedded scaffolds such as instructional guidance, immediate feedback, performance grading and quizzes.



Video: VR-IVPB Medication Administration Collaborative Practice



School of Information Science
& Learning Technologies
University of Missouri

Methods

Design

- Post-intervention mixed-methods pilot study

Participants

- 26 nursing students in the peered practice condition

Procedure

- Demographic survey
- Orientation video and headset/controller practice
- Individual VR practice
- Collaborative VR practice
- Final independent VR practice
- Total intervention time: about 90 minutes

Data sources

- Self-Assessed Collaboration Skills survey
- Two rounds of semi-structured interviews



(a)



(b)



(c)

Fig. 7 Figures illustrating collaborative VR-IVPB Medication Administration practice: (a) Screenshot from the viewpoint of Nursing Student A interacting with peer Nursing Student B; (b) Nursing Student A observing Nursing Student B's IV pump manipulation in Room 1; and (c) Nursing Student B demonstrating IV pump manipulation to Nursing Student A in Room 2.



Results

Quantitative results

Collaboration quality was highly positive

Overall SACS score

- $M = 6.18 / 7$
- $SD = 0.66$

Subdomains

- Information Sharing: $M = 6.10$, $SD = 0.87$
- Team Support: $M = 6.67$, $SD = 0.65$
- Team Learning: $M = 5.97$, $SD = 0.82$

Response pattern

- 57.7% of responses were Strongly agree
- 14.0% were Agree

Qualitative results

Four major themes

1. **Enhanced confidence and skill mastery:** Students felt more confident with peer reassurance and immediate correction
2. **Collaborative communication and peer feedback:** Students learned from each other through constant dialogue and feedback
3. **Reduced cognitive load and improved task management:** Working with a peer made the procedure feel less stressful and more manageable
4. **Enhanced clinical readiness and performance:** Students felt better prepared for the later physical IVPB evaluation



Discussion and Takeaway

Limitations

- Peer interaction supported confidence, communication, and task management
- Collaborative VR may be a promising way to support complex procedural practice in nursing education
- This pilot study suggests strong potential for teamwork-oriented VR training
- Small sample size
- Single institution
- No direct objective skill-performance measures

Next steps

- Larger and more diverse samples
- Objective performance assessments
- Long-term learning outcomes



Thank You!

Yupei Duan: yupei.duan@missouri.edu

Dr. Xinhao Xu: xuxinhao@missouri.edu



School of Information Science
& Learning Technologies
University of Missouri